Walkthrough deck

Color-pattern diversity, *with and without history*

Demo 8 · pavo + phylogeny + measurement error

EEB 187 · Wednesday May 20, 2026 · led by Rosamari

Companion to the 20-slide intro deck.

The hour ahead

three questions stacked on each other

We will ask, for our pavo color-pattern data:

1. **Is there structure in the standing diversity?**
   PCA + bivariate fits (Part 1, ahistorical)
2. **Is that structure real biology — or just shared ancestry?**
   phyloPCA + PGLS + K and λ (Part 2)
3. **Does the structure survive measurement noise?**
   5-image error stack → 100-iter Monte Carlo (Part 3)

Where we left off last week

Frédérich et al. 2026 — the headline result we'll test

Reef-fish color patterns evolve fast — much faster than Brownian motion predicts.

- **Their question** — do color patterns track phylogeny in 5 reef-fish families?
- **Their method** — 30 binary motif columns per species; one multivariate Blomberg's K.
- **Their answer** — weak phylogenetic signal in every family they tested (K ≈ 0.05–0.22).
- **Their interpretation** — convergence is rampant; you can't read color off the tree.

Today we ask the same question with continuous pavo statistics, variable-by-variable, on our own family — and we stress-test it under noise.

Our walkthrough family — Chaetodontidae

19 butterflyfish species · 5 genera · 1 error-species

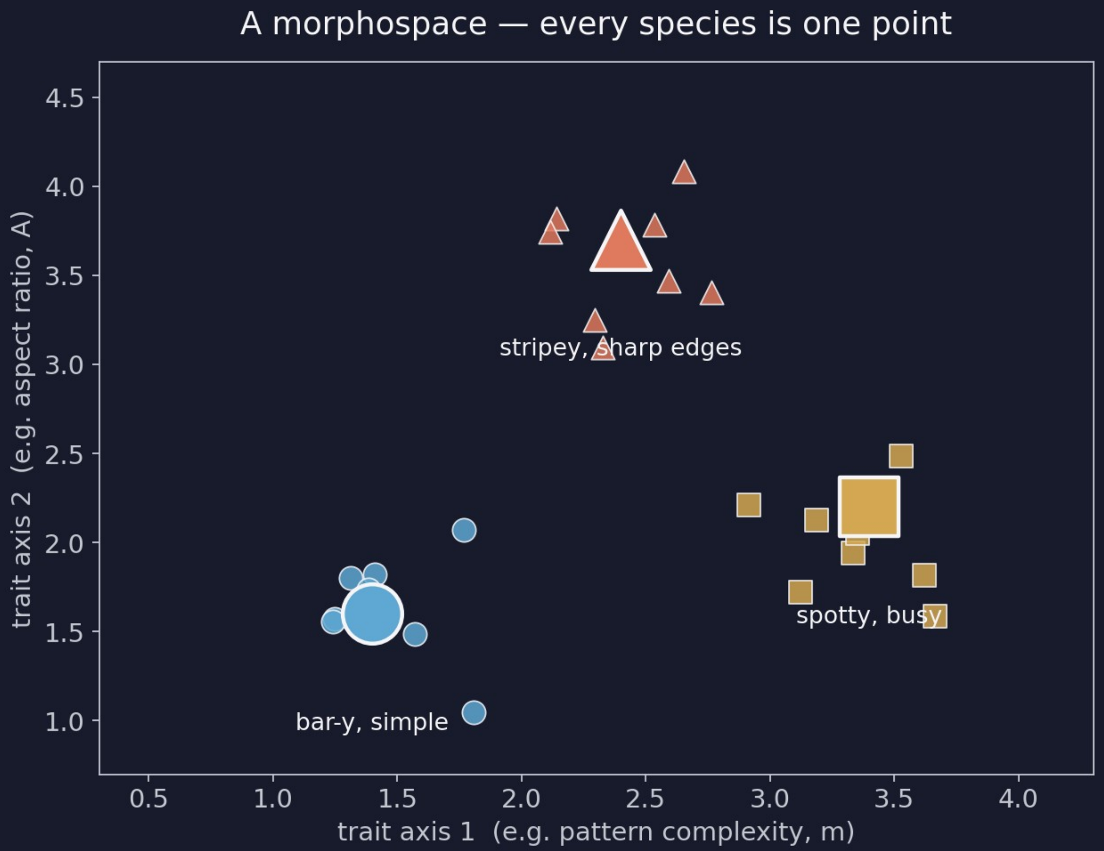
Same code, same plots that every group will rerun on their own family.

- **19 species** — across Chaetodon, Heniochus, Roa, Amphichaetodon, Prognathodes
- **1 image per species** — exemplar.png — chosen by the picker for lateral, in-focus view
- **1 error-species** — Heniochus_acuminatus, photographed 5 times — the noise stack for Part 3
- **6 pavo descriptors** — m, A, Jc, Jt, m_dS, m_dL (full-rank, no algebraic dependencies)
- **4 FishBase traits** — Troph, max depth, max length, vulnerability
- **1 pruned chronogram** — tip labels matched to image directories by the picker

What is a morphospace?

the foundational idea behind today's analysis

Every species becomes one point in a space whose axes are trait values.



- Axes = traits (or combinations of traits)
- Points = species
- Close = similar pattern
- Far = different pattern
- Clusters = groups of species that look alike for the same reason

The 6 pattern descriptors we keep

pavo's classify() + adjacent() — six full-rank columns

m complexity (0 → 1 = mosaic)

A directional aspect

> 1 bar-y · < 1 stripe-y

Jc palette use

how much of the palette

the pattern uses

Jt transition variety

color-pair edges realized

m_dS chromatic boundary

sharp color contrasts

m_dL luminance boundary

sharp dark-light steps

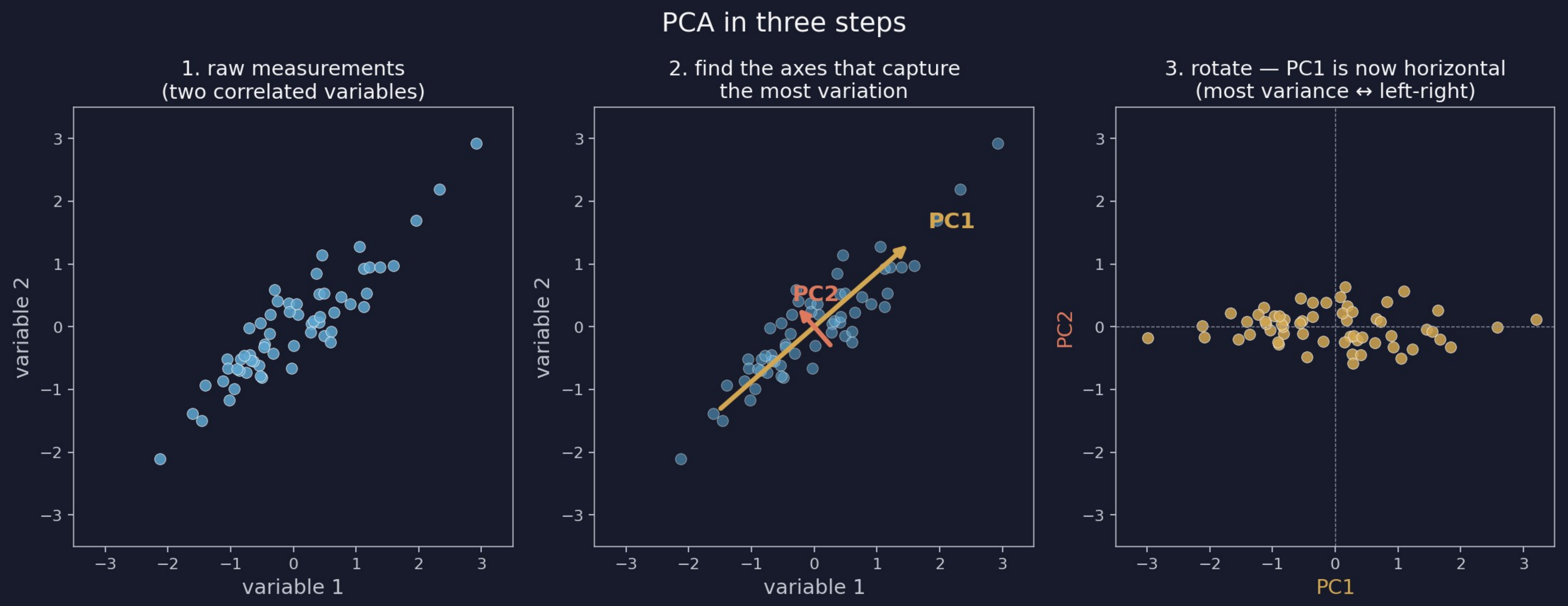
Six dimensions → two — why we need PCA

You cannot plot 6 axes on a screen. PCA finds the 2 axes that capture the most variation.

- **The problem** — with 6 pavo variables, every species is a point in 6-D — invisible.
- **Naïve fix** — plot pairs of variables (Demo 6's pairs() plot). 15 sub-plots; brittle.
- **PCA** — rotates the 6-D cloud so the first axis captures the most variance, the second captures the most of what's left, and so on.
- **Why centre + scale** — our variables are on different units (proportions vs probabilities vs ratios). Without scaling, the variable with the biggest raw spread dominates the axes.
- **How many PCs to read** — look at the scree — it tells you when extra PCs stop carrying real signal.

PCA in three steps

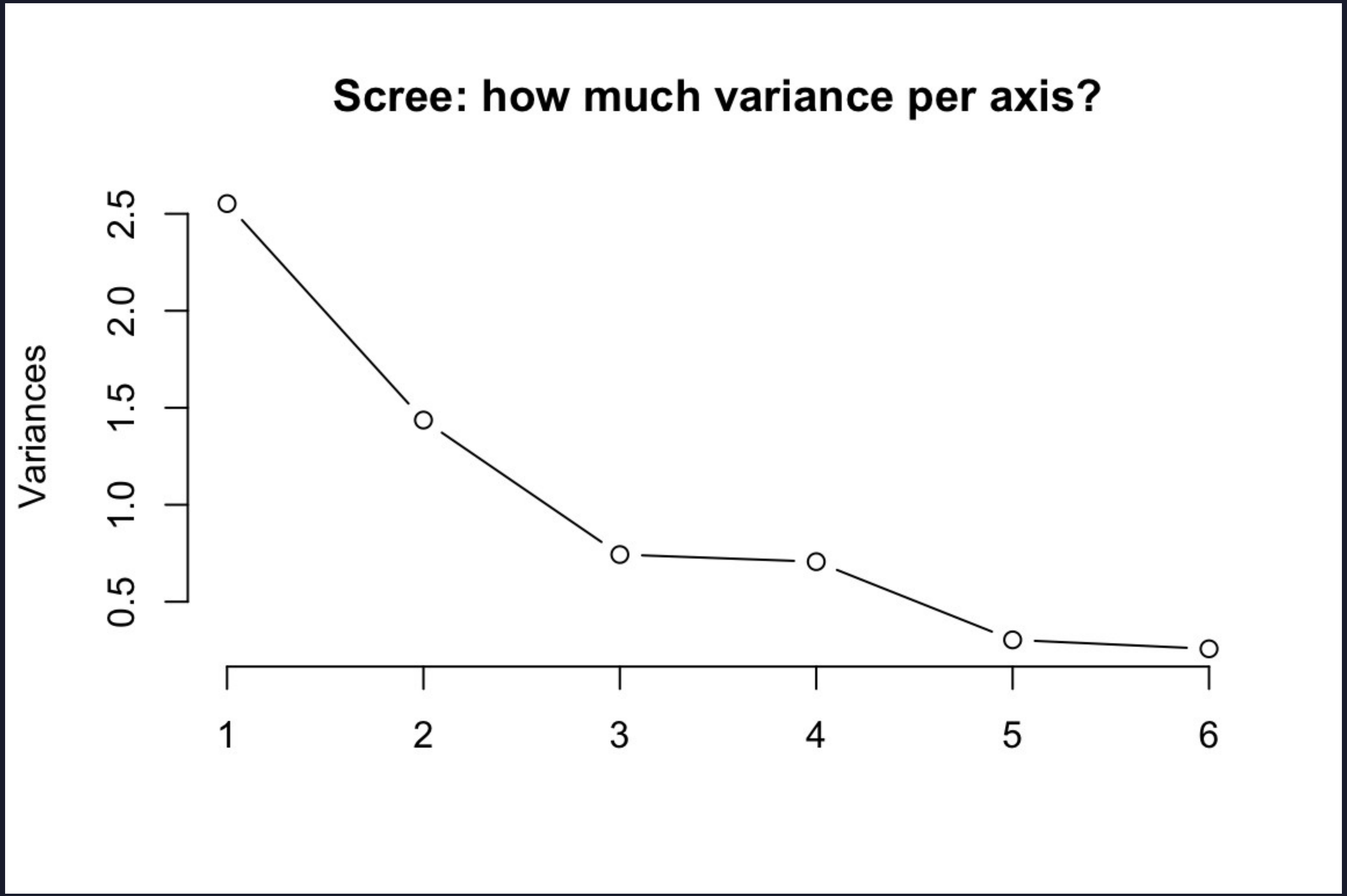
raw scatter → find the axes → rotate



Step 3 is the picture you'll read. PC1 horizontal, PC2 vertical, every species one point.

Reading the scree plot

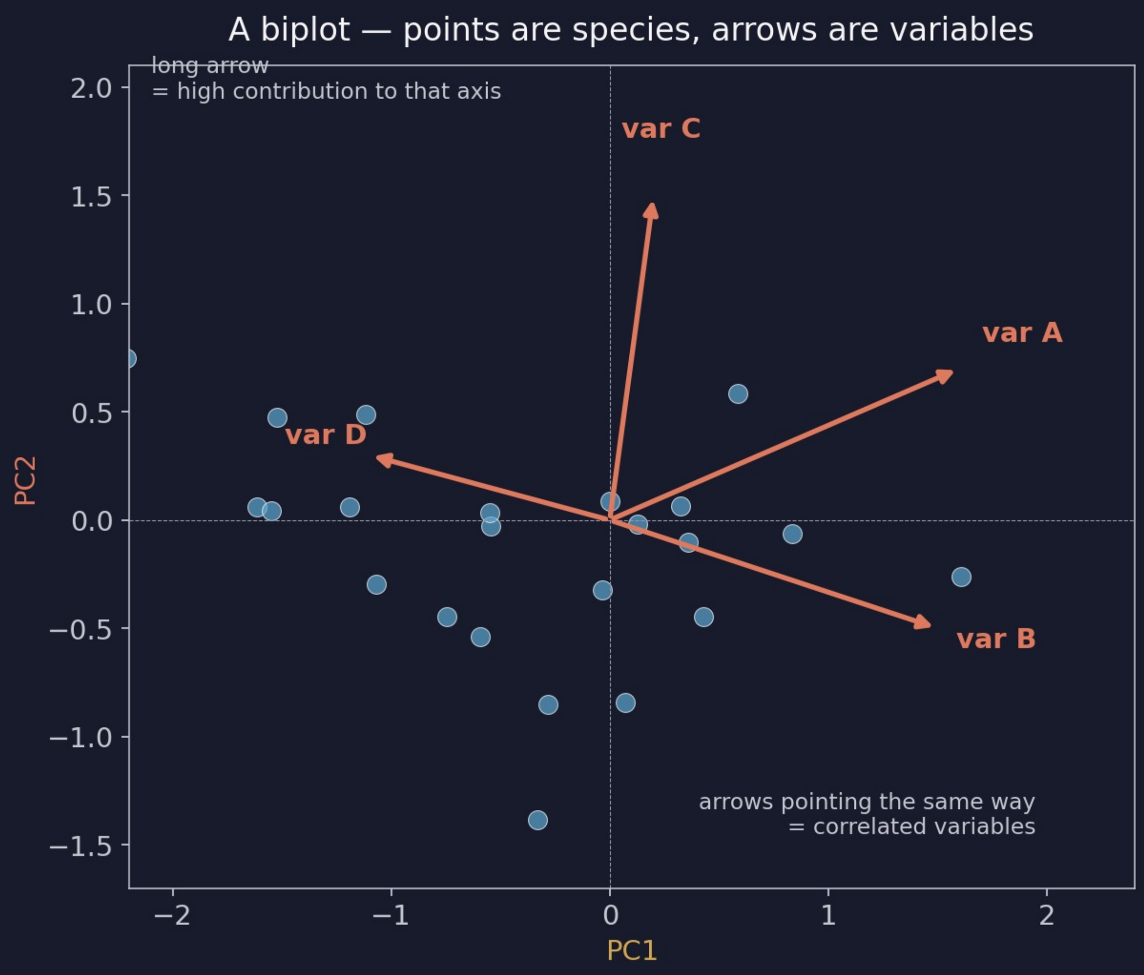
how much variance does each PC actually carry?



- **PC1 always wins** — the math forces it.
- **Question:** — by how much does it win?
- **Elbow** — the bend where PCs stop carrying real signal.
- **Our data** — PC1 ≈ 42 %, PC2 ≈ 24 % — together about two-thirds of the variance.
- **Verdict** — two axes is a usable summary.

Reading the biplot — arrows are loadings

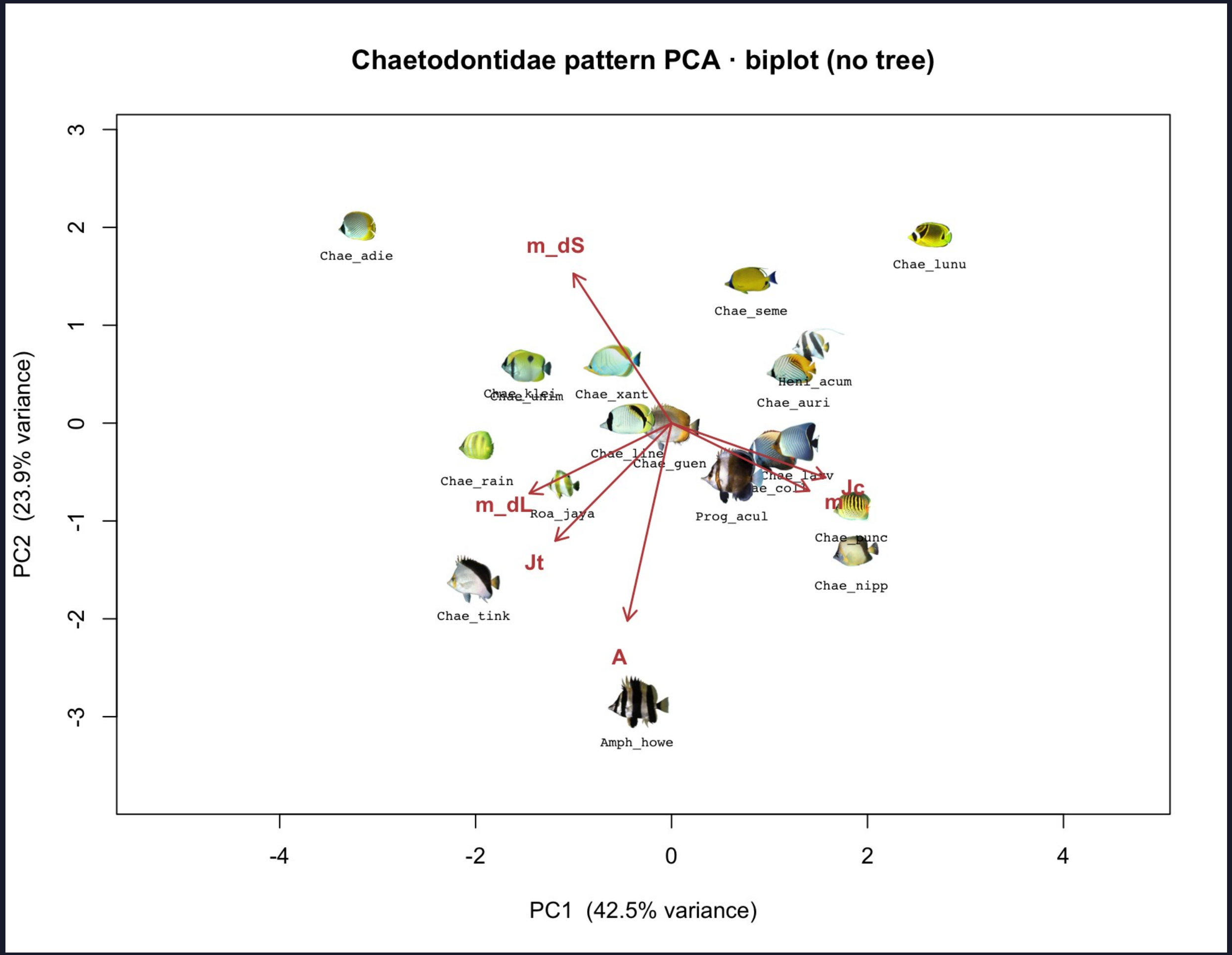
points are species; arrows are pavo variables



- **Long arrow** — high contribution to that axis.
- **Short arrow** — low contribution.
- **Arrows same direction** — those variables are correlated.
- **Arrows opposite** — they trade off.
- **Arrow perpendicular to an axis** — carries no information on that axis.

The Chaetodontidae biplot — our actual data

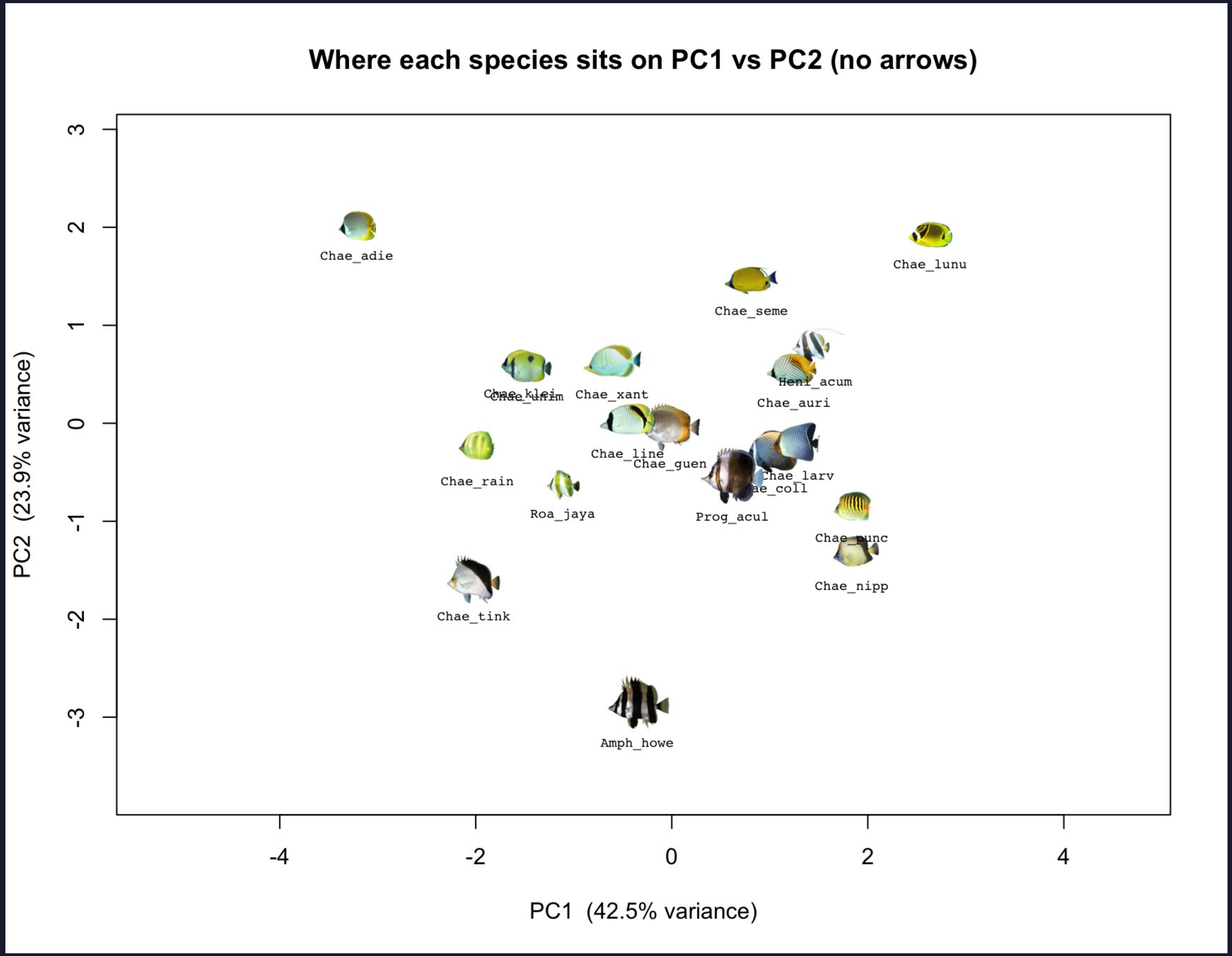
PC1 ≈ 42.5 % · PC2 ≈ 23.9 % · arrows are pavo variables



PC1 driven by m + Jc (busy patterns). PC2 driven by A + Jt + boundary metrics.

The Chaetodontidae morphospace

every thumbnail is one species; same pavo stats → same neighborhood



Does ecology predict pattern?

the question §1.8 of the worksheet sets up

If we tell you how a fish lives, can we predict what its color pattern looks like?

- **Hypothesis** — depth, diet, body size shape the visual world a fish lives in, so its pattern should evolve to match.
- **If true** — ecology variables predict pavo PC1 (or specific pavo variables).
- **If false** — color and ecology are decoupled — pattern is driven by other things (mate choice, lineage history, sensory drive in non-obvious ways).
- **Frédérich 2026 says** — for these families, color evolves fast enough that ecology and color are mostly decoupled. Today we test that on Chaetodontidae.

The two named tests in Part 1

ahistorical — we ignore the tree on purpose, just to see what's there

- **m ~ Troph** — do herbivores have less busy patterns than piscivores?
- **A ~ log_depth** — does pattern orientation change with depth and the light environment?

These are pretending each species is an independent observation. Part 2 will redo them with phylogenetic correction.

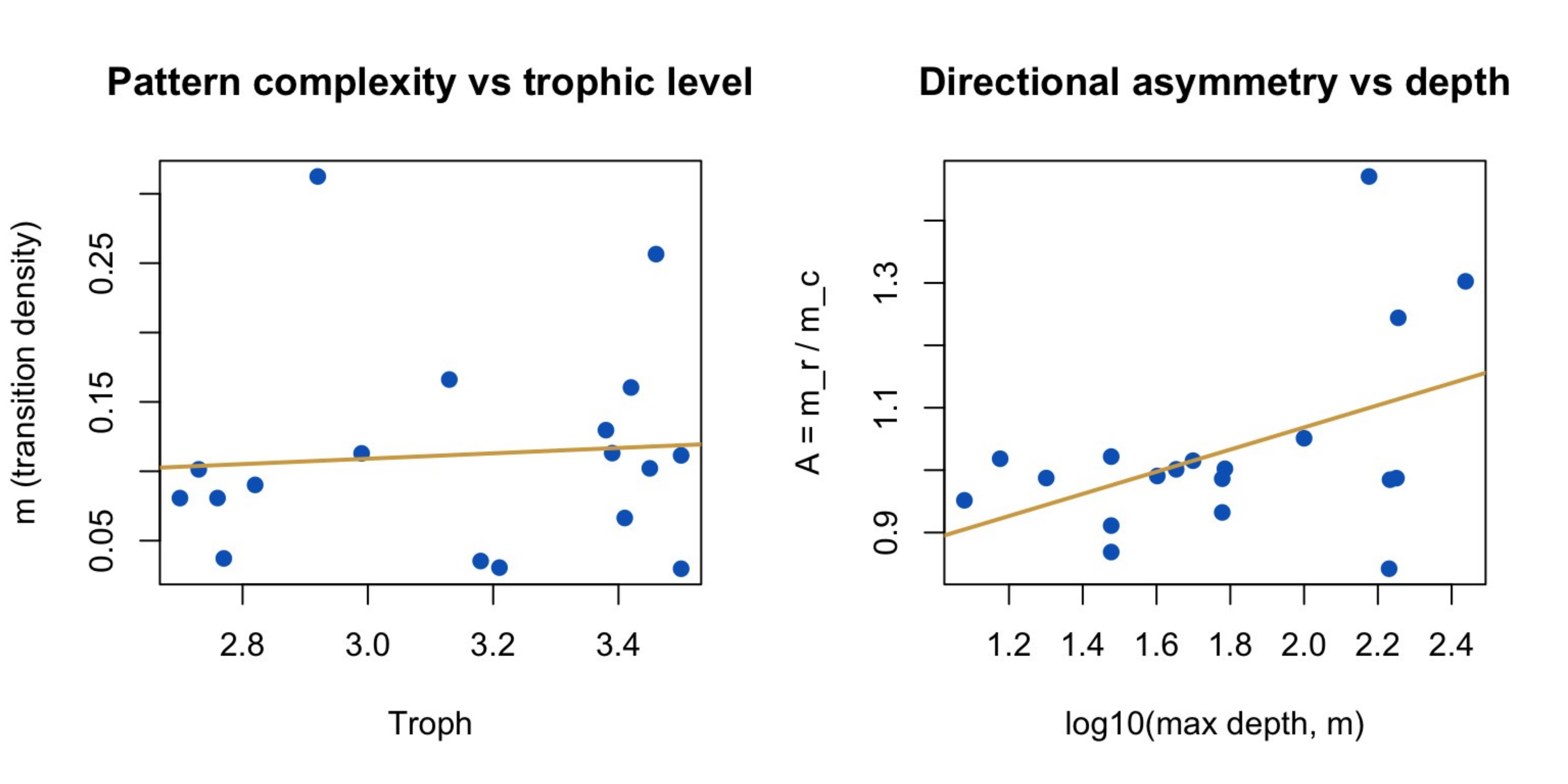
Workflow: lm() in R, exactly the same as Demo 6.

Output: slope, intercept, p-value, R-squared.

Read: is the slope distinguishable from zero? How much variance does it explain?

Ecology scatter — what we see ahistorically

both slopes weak in 19 butterflyfish species

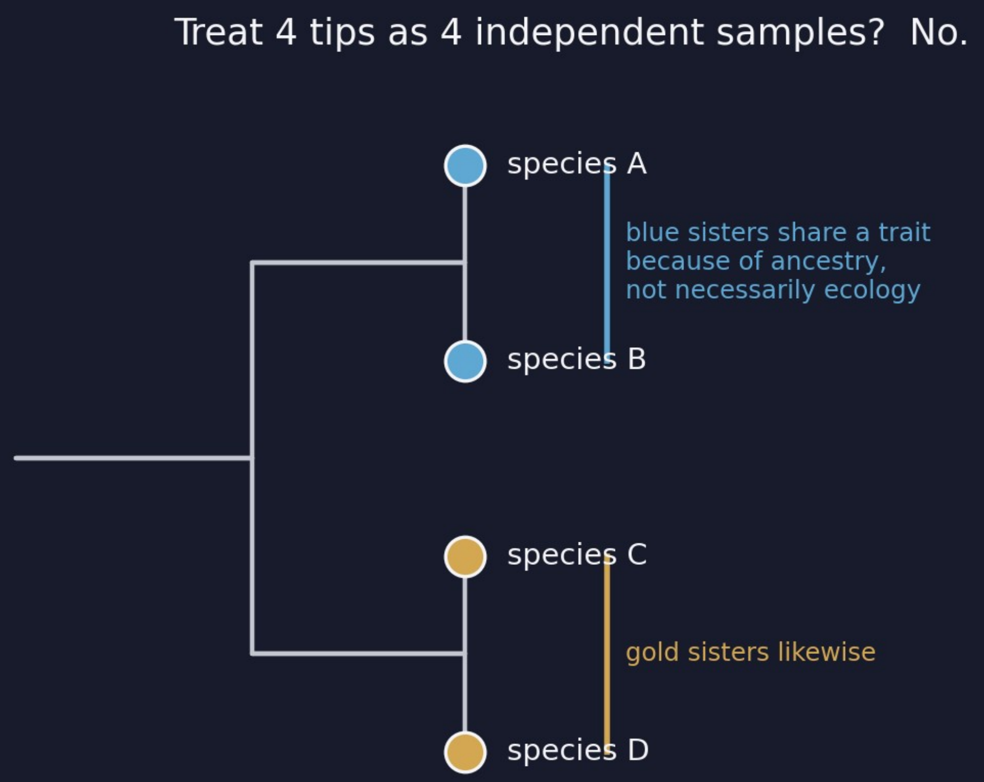


- **Trophic level → m** — $R^2 < 0.1$, $p > 0.3$.
- **Depth → A** — also weak.
- **Caveat** — small sample. Bigger trees may show more.
- **Bigger caveat** — these tests don't know about phylogeny.
- **Concept block 3** — fixes that.

Why phylogeny matters

shared ancestry breaks the independence assumption

Closely related species share trait values because they share genes — not necessarily because of ecology.



- **Naïve view** — 4 tips = 4 independent samples.
- **Reality** — 2 sister pairs sharing ancestors.
- **Consequence** — the apparent correlation between blue and gold traits is partly inherited, not earned.
- **Fix** — downweight the sisters when fitting.

Two corrections we will apply

one for multivariate axes, one for bivariate fits

- **phyloPCA** — PCA that uses the tree's variance-covariance matrix to weight species.
- **PGLS** — generalized least squares where residuals can correlate via the tree — the bivariate-lm analog of phyloPCA.
- **Plus** — Blomberg's K and Pagel's λ on every pavo variable, to ask which traits carry phylogenetic signal at all.

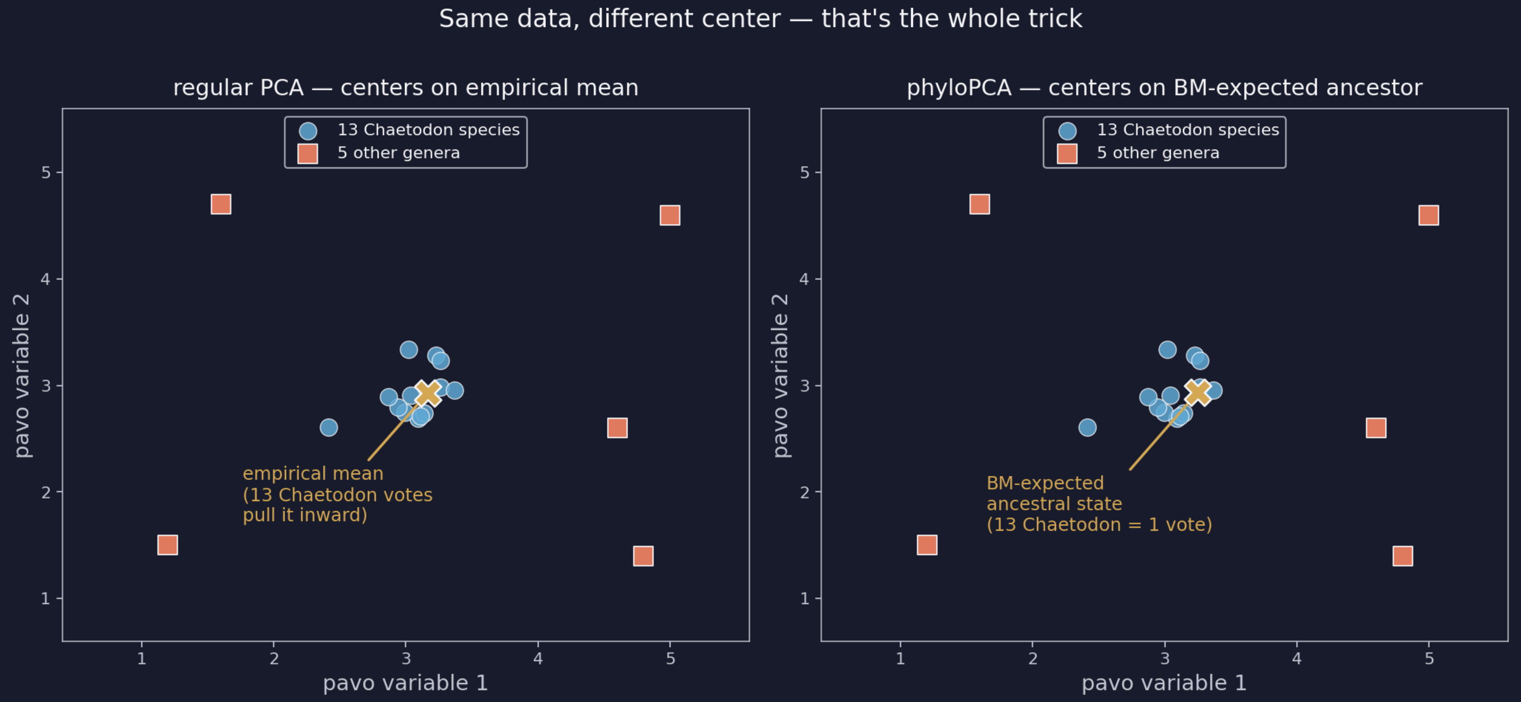
All three use the same math machinery: invert the tree's variance-covariance matrix and downweight non-independent species.

If the tree is a flat star (no internal structure) every correction reduces to the ahistorical version.

phyloPCA — what it does that regular PCA doesn't

same axes, different centering

Regular PCA centers on the empirical mean. phyloPCA centers on the BM-expected ancestral state.



13 bar-y Chaetodon = 13 votes for 'bar-y is one direction' under regular PCA. Under phyloPCA, 1 vote (the ancestor).

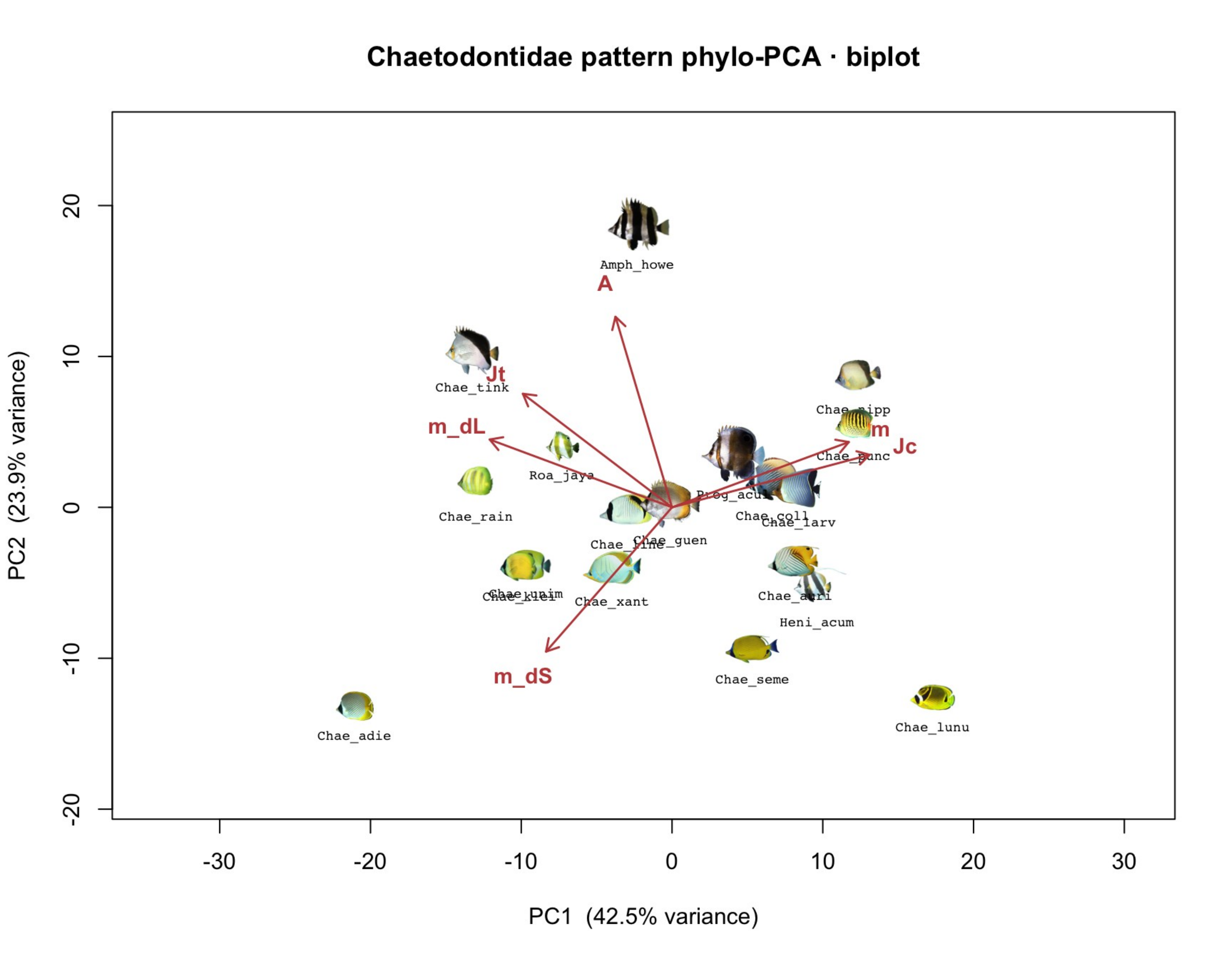
phyloPCA — unpacked

the consequence for loadings

- **Regular PCA** — rotates the cloud of 19 species treated as 19 independent points.
- **phyloPCA** — rotates the cloud of independent evolutionary changes — sister species get pulled toward each other in the variance-covariance step.
- **So the new axes** — describe directions of evolutionary change in pattern space, not just directions of current correlation.
- **Loadings differ from regular PCA** — when a variable's apparent strength came from a clade of similar species. phyloPCA strips out that inflation.
- **Same math as PGLS** — both invert the tree's variance-covariance matrix.

phyloPCA biplot for Chaetodontidae

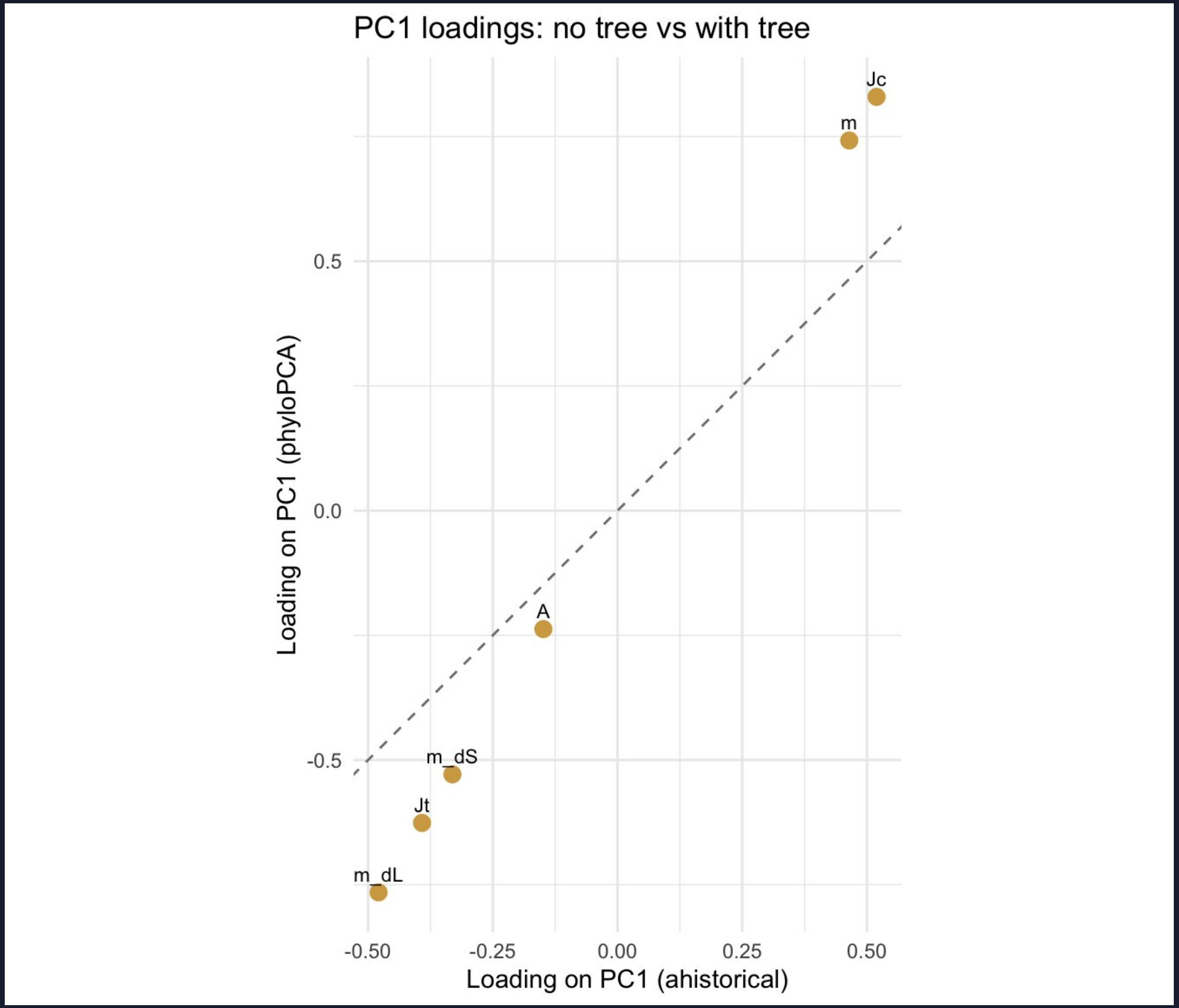
the Chaetodon cluster vs the outer genera



- **Cluster near origin** — the dense Chaetodon clade.
- **Splayed outward** — Roa, Heniochus, Amphichaetodon, Prognathodes.
- **This shape** — is the multivariate fingerprint of phylogenetic structure in pattern space.
- **Compare to slide 13** — regular PCA spread Chaetodon out; phyloPCA pulls them together.

Did phylogeny change which variables matter?

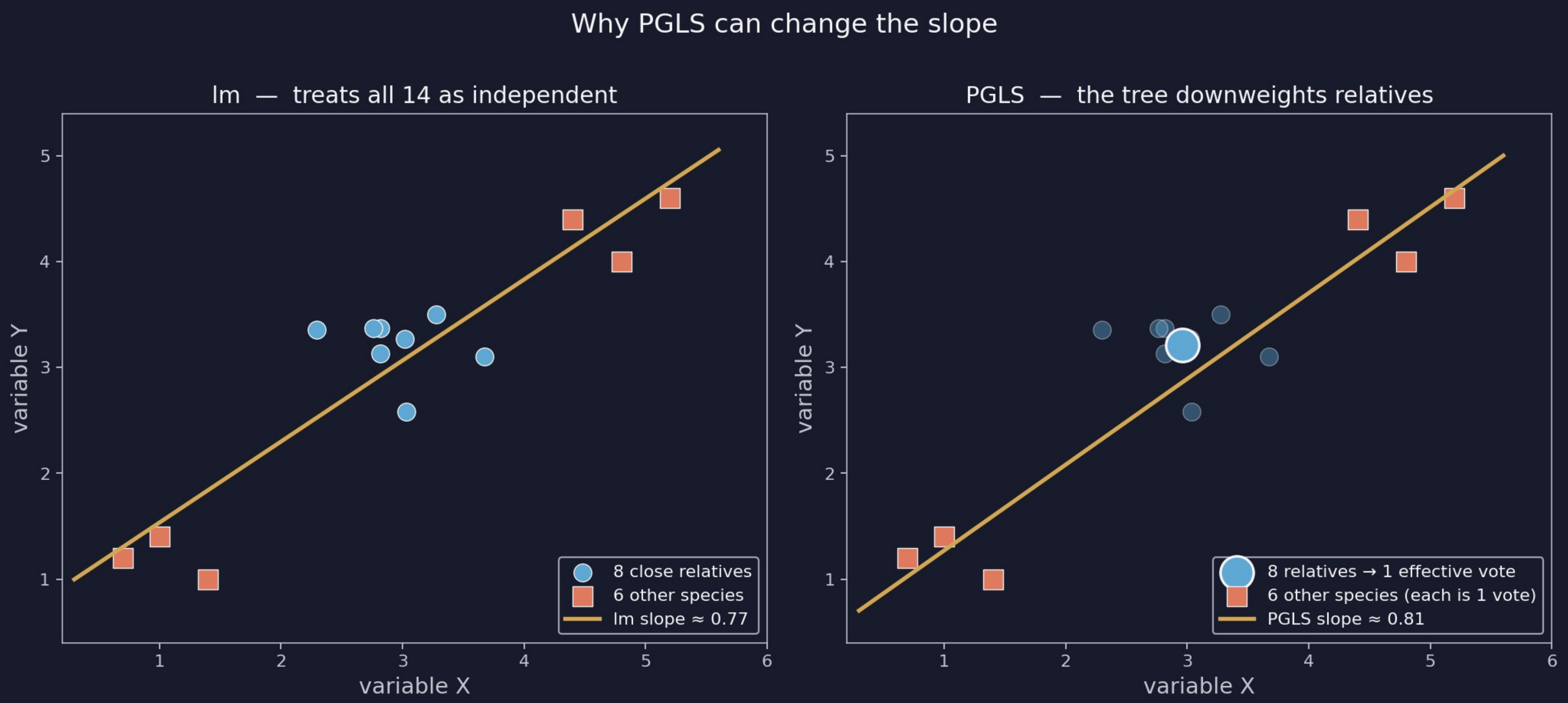
PC1 loading — regular PCA vs phyloPCA



- **On the dashed y = x line** — phylogeny didn't change that variable's loading.
- **Above the line** — phyloPCA increased the loading — ancestry was masking it.
- **Below the line** — phyloPCA decreased the loading — ancestry was inflating it.
- **Read variable-by-variable** — not as a single number.

PGLS — bivariate fits with phylogenetic correction

same idea as phyloPCA, but for one X-Y relationship at a time

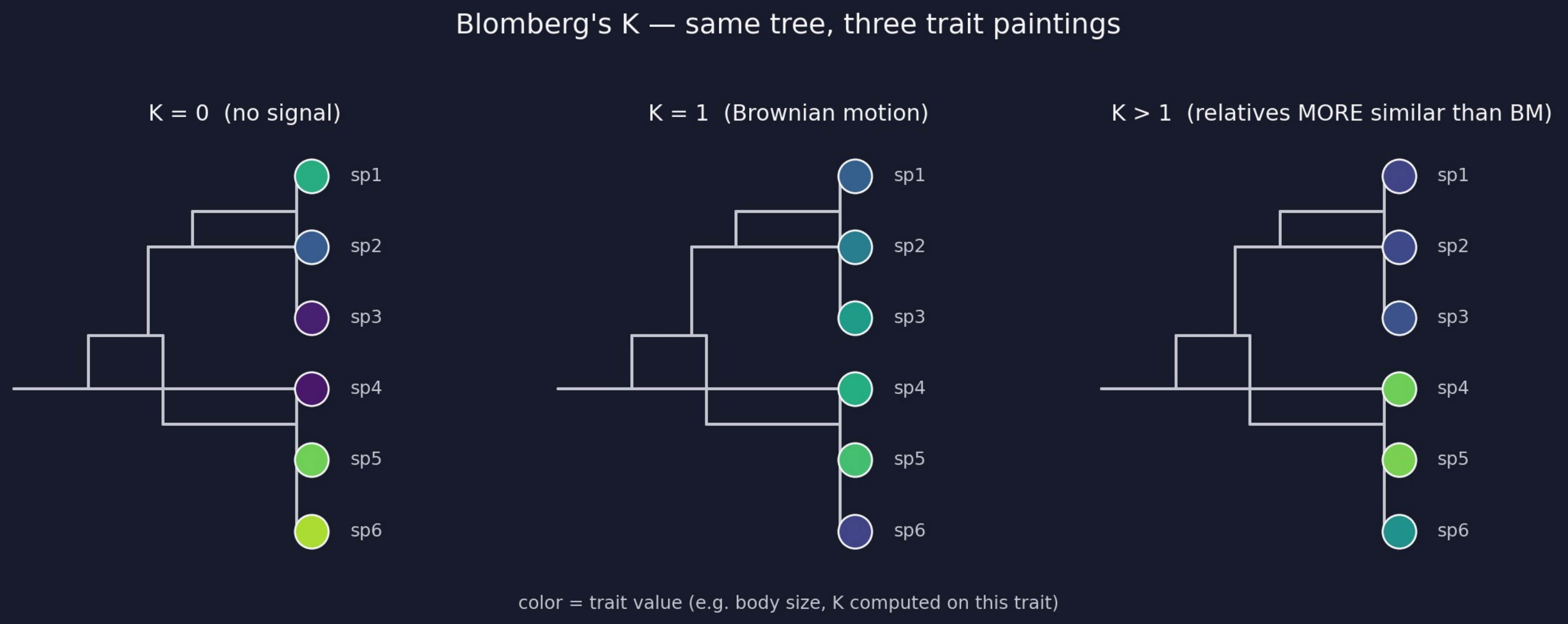


Same two fits as §1.6 — A ~ m and m_dL ~ m_dS — but the tree downweights non-independent species.

Blomberg's K — measuring phylogenetic signal

one number per trait

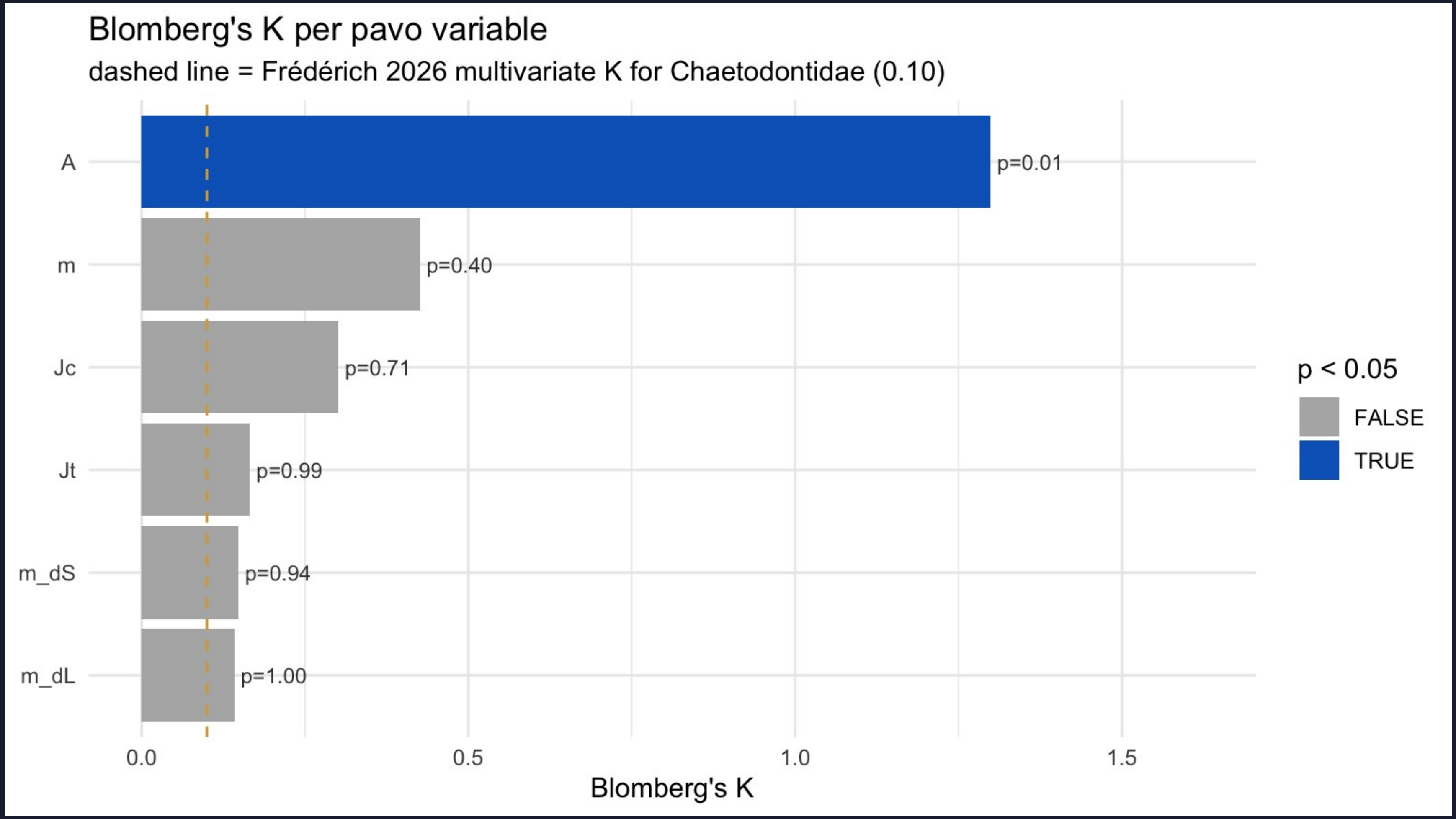
K compares how similar relatives actually are to how similar Brownian motion predicts they'd be.



- **K = 0** — trait randomly distributed on tree — relatives no more similar than distant species.
- **K = 1** — trait matches Brownian-motion expectation exactly.
- **K > 1** — relatives MORE similar than BM predicts — active conservation.

K per pavo variable in Chaetodontidae

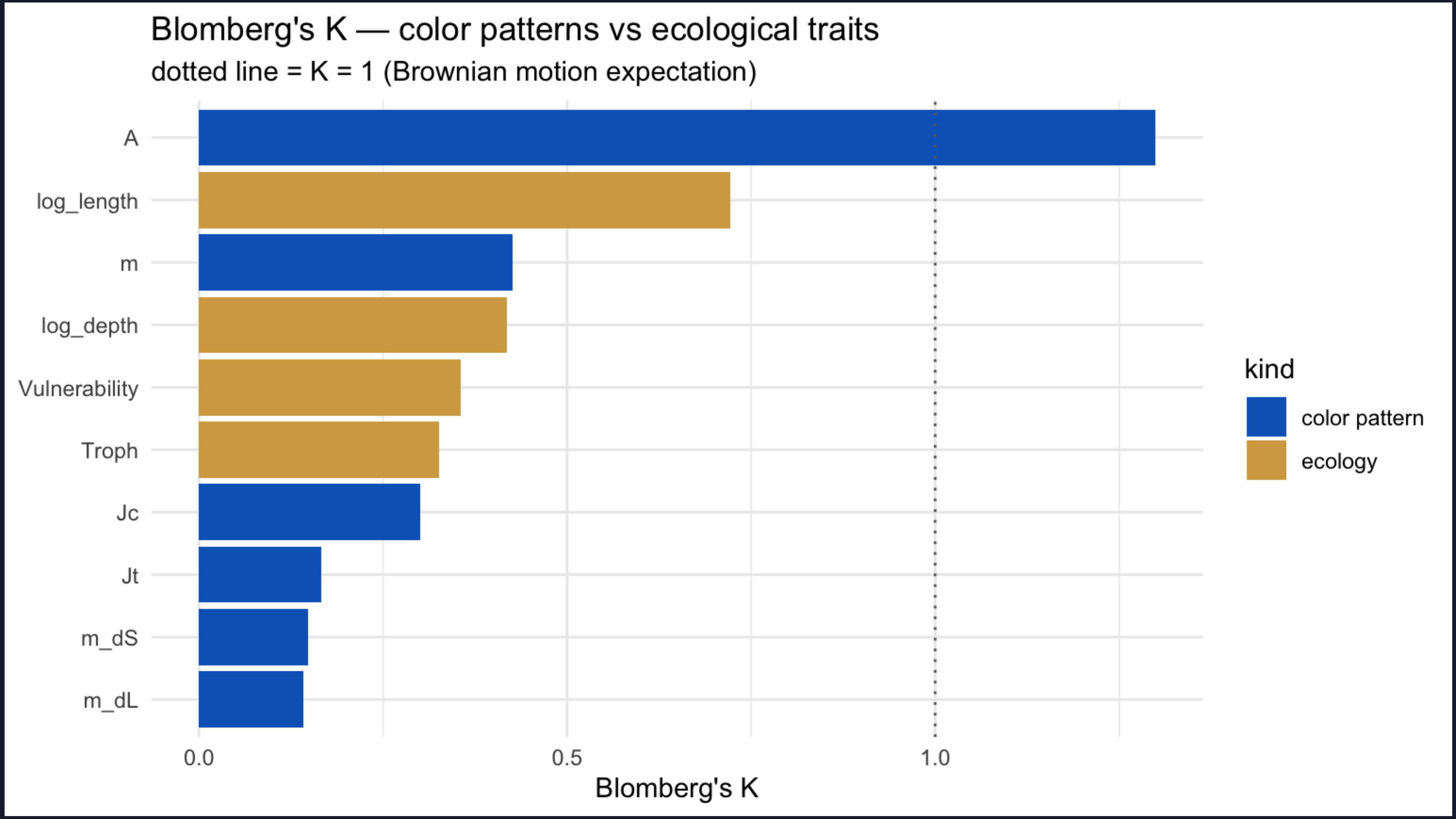
one bar per variable — colored if $p < 0.05$



- **A → K ≈ 1.3** — p ≈ 0.01. Relatives MORE similar than BM.
- **The other 5** — K between 0.15 and 0.45, non-significant — labile.
- **Take-home** — pattern orientation is conserved in butterflyfishes; complexity, palette, and boundaries are not.
- **Frédérich landmark** — dashed line at K = 0.10 — not a target our bars should hit.

Color signal vs ecology signal

do close relatives share how they live more than how they look?



- **Ecology bars above color bars** — ecology is more phylogenetically conserved than pattern.
- **Interpretation** — close relatives are tied by how they live more than how they look.
- **In our 19-tip sample** — the bars overlap — sample size is the limiting factor.
- **Your family** — may show the pattern more sharply if it has deeper splits.

Part 2 take-home for Chaetodontidae

the result we'll stress-test in Part 3

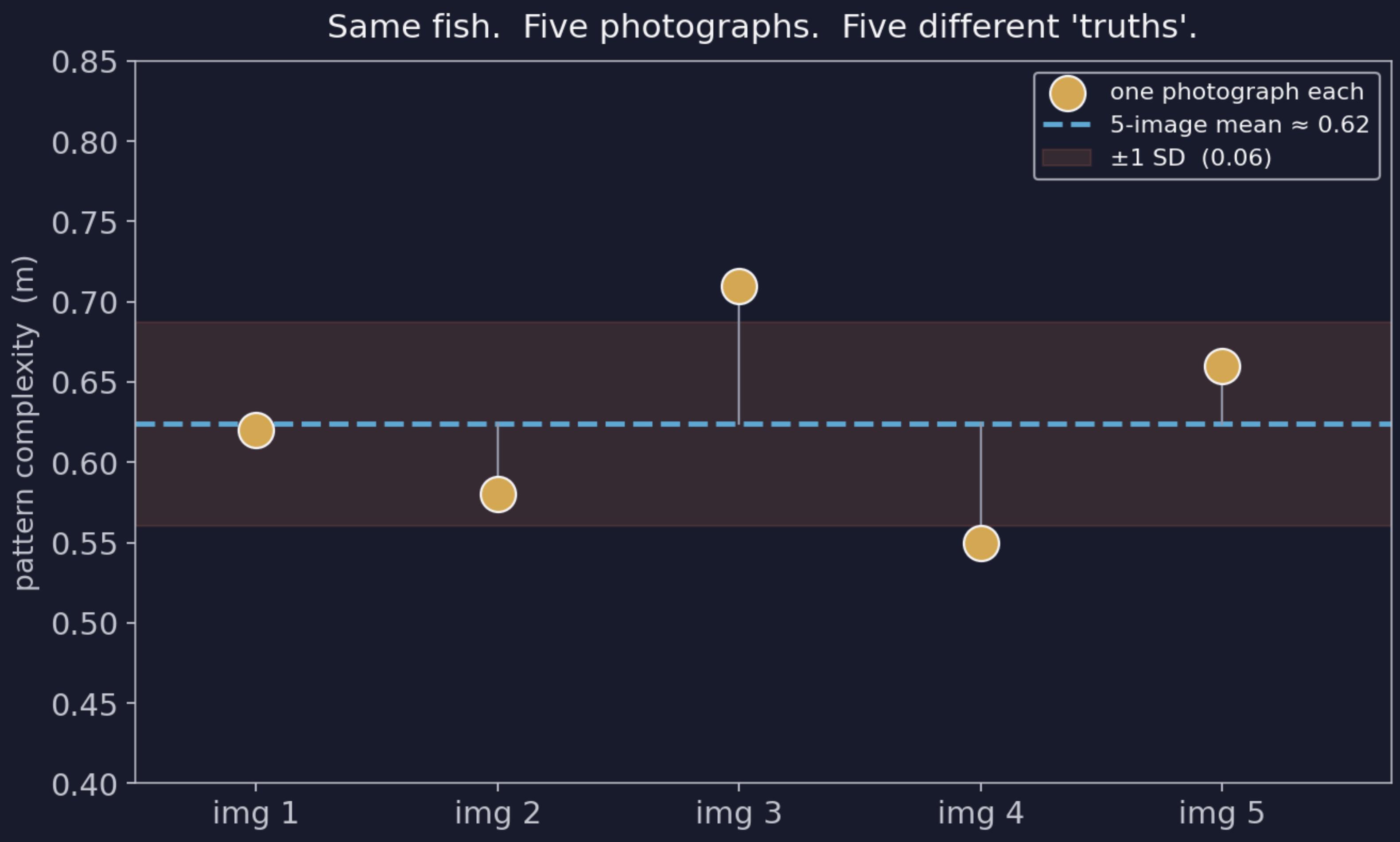
Pattern orientation is conserved within butterflyfishes; everything else is evolutionarily labile.

- **Conserved** — A (pattern aspect ratio) — bar-y stays bar-y, stripe-y stays stripe-y.
- **Labile** — m, Jc, Jt, m_dS, m_dL — complexity, palette use, transition variety, and both boundary strengths.
- **Bivariate story** — after PGLS, m_dL ~ m_dS gains a significant positive slope; A ~ m stays flat.
- **Matches Frédérich** — qualitatively — color patterns largely evolve fast. The exception is A.

Part 3 asks: how much of that survives noisy measurements?

Why a single number is a lie

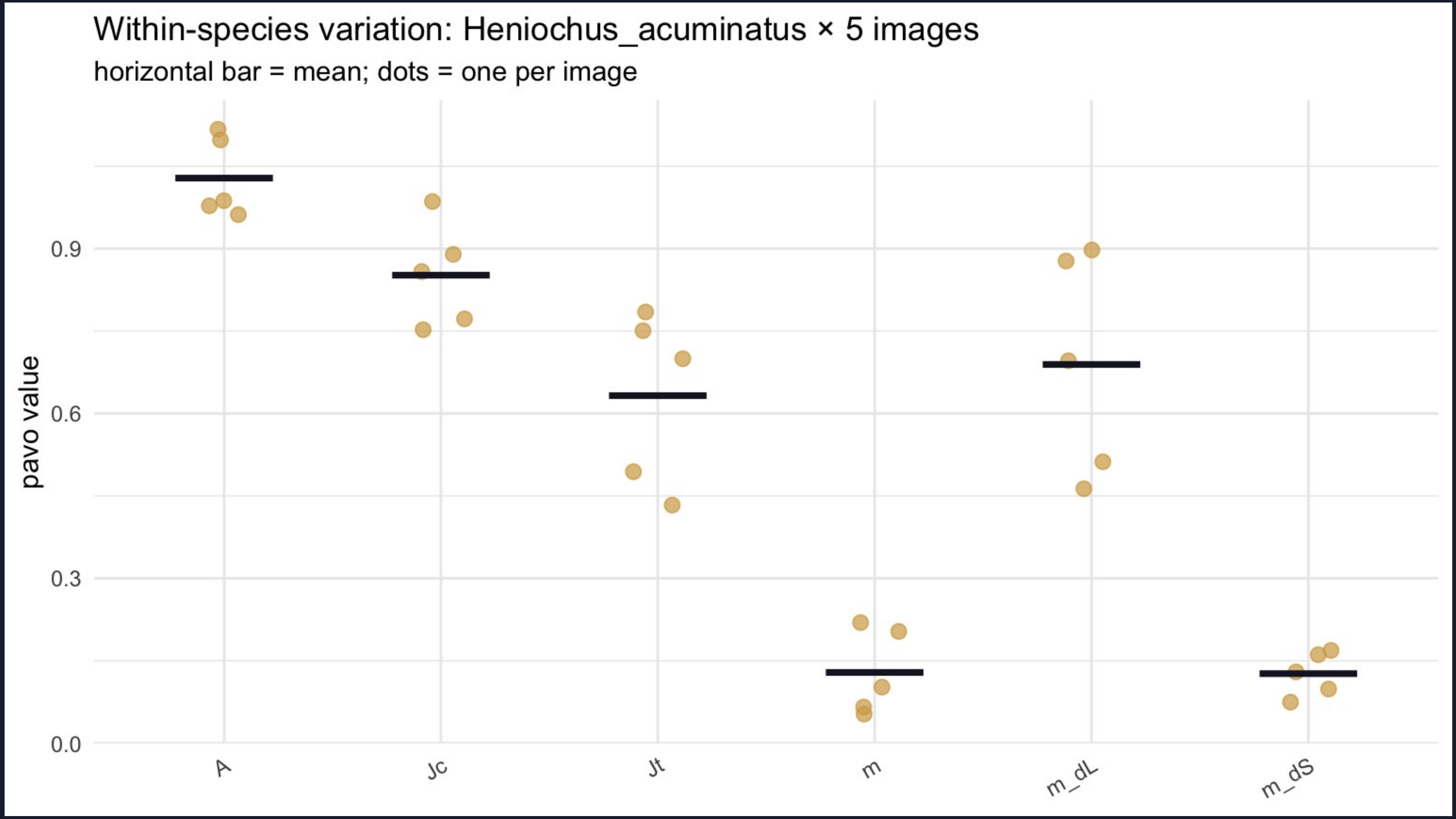
every pavo statistic came from one photograph



- **Same fish** — Same species, same individual.
- **Different photos** — Different angle, light, framing.
- **Different pavo values** — the 'truth' is a distribution.
- **Part 3** — asks how much that distribution matters.

The 5-image error stack

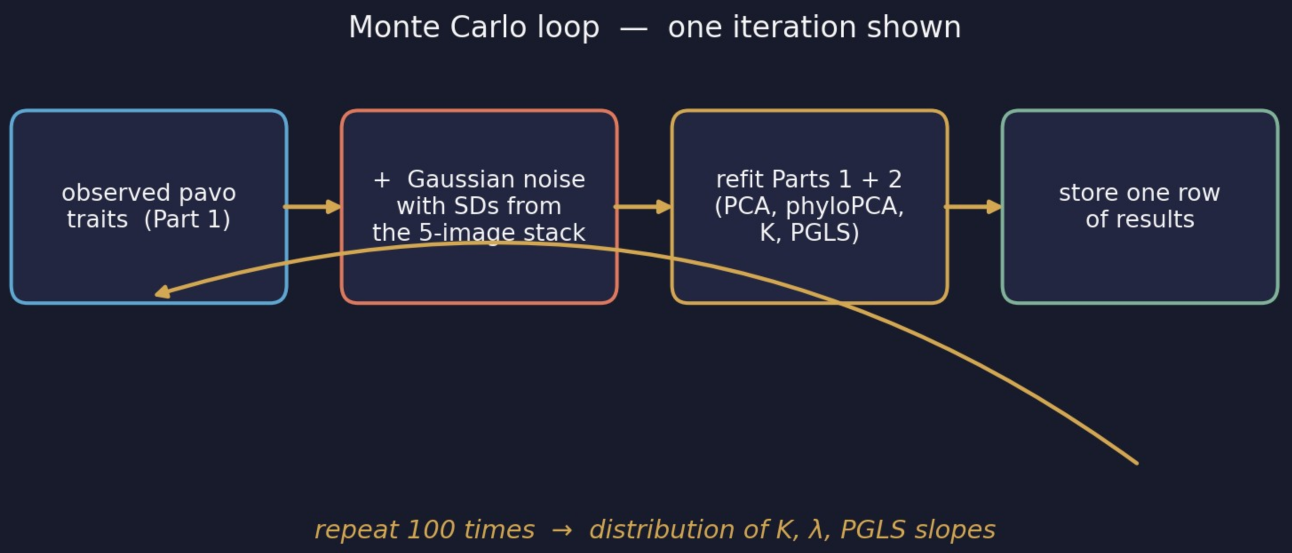
one species photographed 5 times → 5 pavo values per variable



- **Dots** — one pavo value per photograph.
- **Bar** — the 5-photo mean.
- **Tight clusters** — pavo is reliable for that variable.
- **Wide spread** — that variable's Part-2 result is fragile.
- **Simplifying assumption** — treat the SD from this one species as the SD for every species.

Monte Carlo — re-run everything under noise

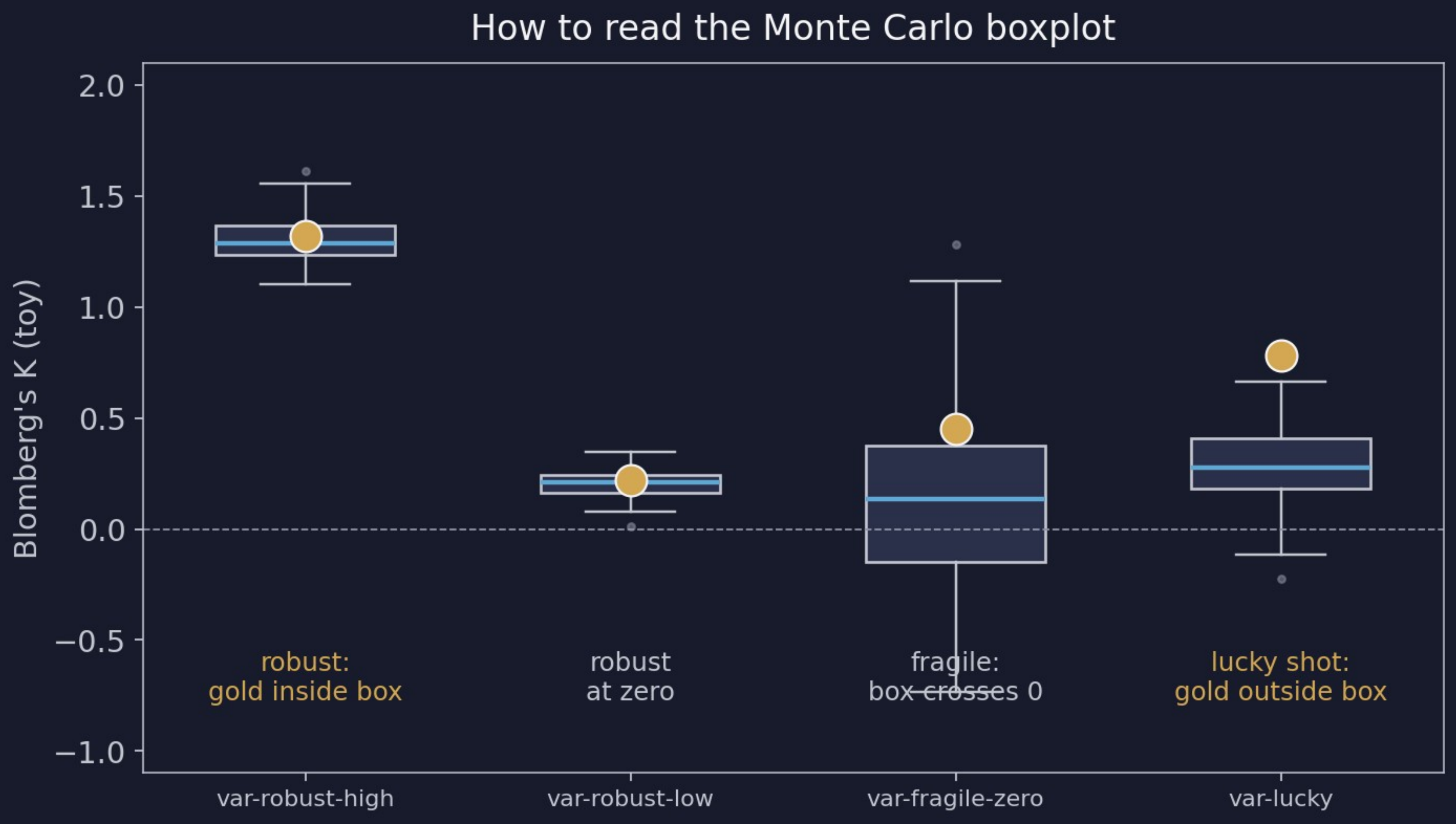
100 iterations, each adding fresh Gaussian noise



- **Each iteration** — add Gaussian noise with the observed SDs to every tip; refit everything.
- **Store** — K, λ, PGLS slopes.
- **After 100 iterations** — you have 100 K values per pavo variable — a distribution, not a point.
- **Time cost** — ~1.5–2 min for n_iter = 100, ~3–5 min for n_iter = 200.

Reading the Monte Carlo boxplot

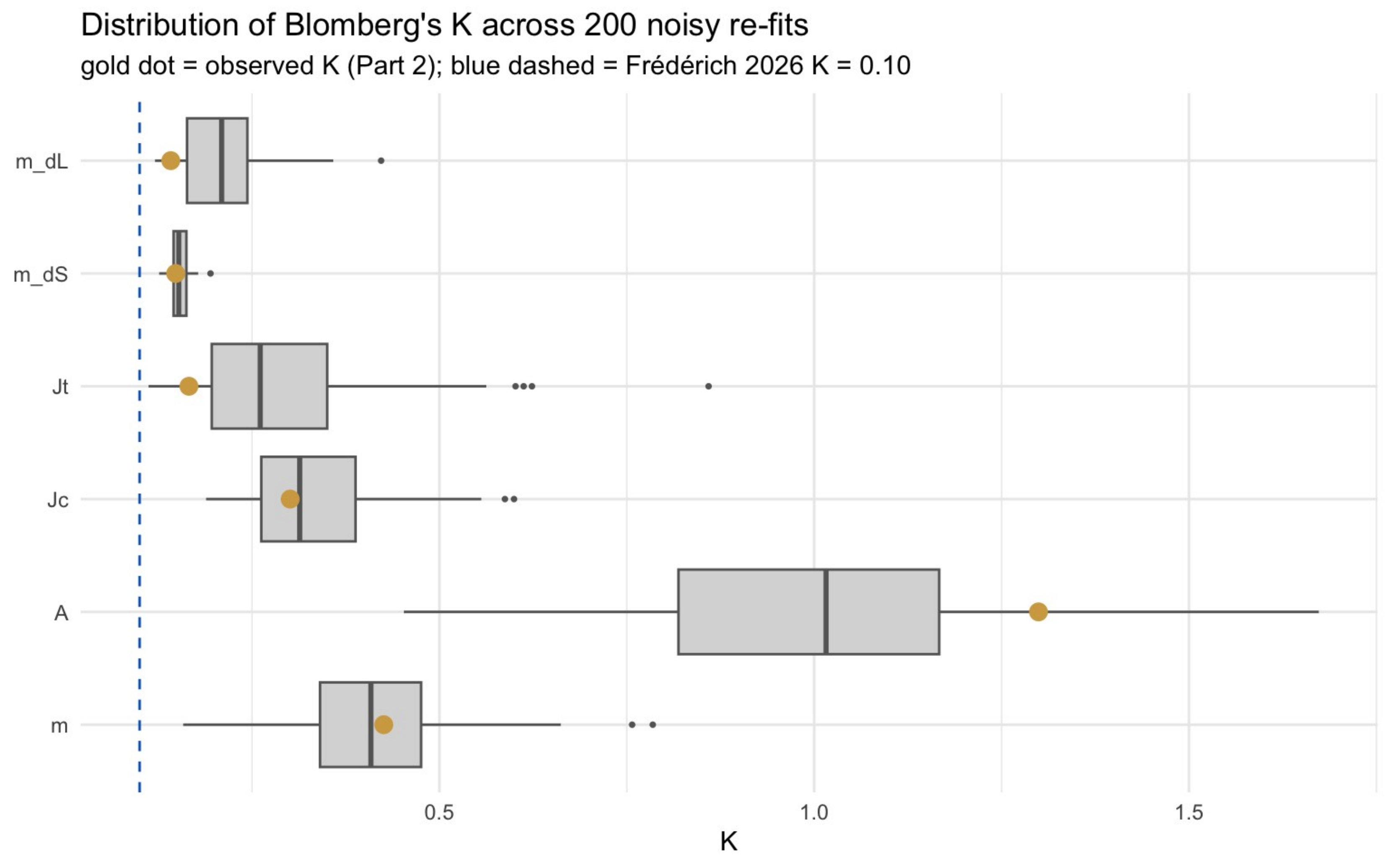
the gold dot is the observed value from Part 2



- **Gold inside the box** — robust — your Part 2 conclusion holds.
- **Box centered at zero** — no signal — fine if that was the claim.
- **Box straddles zero** — fragile — can't distinguish from no signal.
- **Gold outside the box** — you got lucky — a different photo would have given a different K.
- **For PGLS slopes** — the test is whether zero lies inside the 95 % CI.

Chaetodontidae K under noise

100 Monte Carlo re-fits · gold dot = observed (Part 2)



- **A's K stays high** — robust under noise — pattern orientation really is conserved.
- **Other 5** — boxes hover near zero — consistent with the non-significant point estimates.
- **Verdict** — the headline result from Part 2 survives realistic measurement error.

Part 3 take-home

what survives in Chaetodontidae

Once we admit one photograph isn't the truth, only A's signal still holds. Everything else stays labile.

- **Robust under noise** — A — high K, tight MC distribution, gold dot inside the box.
- **Robust at no-signal** — m, Jc, Jt, m_dS, m_dL — boxes near zero, gold dot inside.
- **No PGLS slope flips** — A ~ m stays flat under noise; m_dL ~ m_dS stays positive.
- **Lesson for your family** — if your headline relies on a variable with wide MC spread, soften the claim.

Now you build your own bundle

15 min on FishView · same login as Demo 7

Pick ≥ 15 species across ≥ 5 genera from your family

Star one exemplar image per species

Mark ONE species as the error-species → tick 5 images for it

Hit Export. Drop the ZIP next to your Chaetodontidae bundle.

The choices you make here ARE the analysis.

Frédérich-paper species are starred for Acanthuridae, Chaetodontidae, Pomacanthidae. The 4 Labrid groups + Balistoidea pick cold — your K estimates are the first for your clade.

What you turn in

BruinLearn · end of day Wednesday

1. Part 1 PCA scatter — your family
2. Part 2 K plot + one-line interpretation per pavo variable
3. Part 3 Monte-Carlo boxplot on K (or one PGLS slope)
4. One paragraph (≤ 250 words) — branch by family:

If your family is in Frédérich (Acanthuridae, Pomacanthidae):

compare your per-variable Ks to their published multivariate K.

If your family is NOT in Frédérich (4 Labrid groups, Balistoidea):

these are the first reported Ks for your clade — describe the pattern.

Tonight

prep page · ~20 minutes

1. Install R packages (one chunk on the prep page)
2. Download the Chaetodontidae walkthrough bundle
3. Run the readiness check until it prints PASS
4. Confirm your Demo 7 FishView login still works

michaelalfaro.github.io/eeb187-demos/demo8/demo8-prep.html

See you Wednesday.

Bring questions. We will move slowly.